

1. Administrative & Study Identification

Full Study Title: A Phase 3, Randomized, Double-blind, Placebo- and Active-Comparator-Controlled Clinical Study of Adjuvant V940 (mRNA-4157) Plus Pembrolizumab Versus Adjuvant Placebo Plus Pembrolizumab in Participants With Resected Stage II, IIIA, IIIB (N2) Non-small Cell Lung Cancer **Short Title/Acronym:** A Study of Intismeran Autogene (V940) Plus Pembrolizumab (MK-3475) Versus Placebo Plus Pembrolizumab in Participants With Non-small Cell Lung Cancer (V940-002) / INTERpath-002 **Protocol Number:** V940-002 **NCT Number:** NCT06077760 **Posting ID:** 350988239740467329 **Sponsor Name:** Merck & Co., Inc. (MSD) and Moderna **Trial Status:** Recruiting **Investigational Product:** V940 (mRNA-4157 / Intismeran Autogene) plus Pembrolizumab (MK-3475 / KEYTRUDA®) **ATC Code:** L01FF02 **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/keytruda-epar-product-information_en.pdf **Phase of Development:** Phase III **Medical Monitor:** Sponsor Clinical Operations / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether adjuvant administration of the individualized neoantigen therapy V940 (intismeran autogene) plus pembrolizumab is superior to placebo plus pembrolizumab with respect to disease-free survival (DFS) as assessed by the investigator. **Secondary Objectives:** To evaluate overall survival (OS), distant metastasis-free survival (DMFS), lung cancer-specific survival (LCSS), patient-reported outcomes (quality of life), and safety.

2.2 Background & Rationale While pembrolizumab is indicated as adjuvant monotherapy following resection and adjuvant chemotherapy for certain patients with non-small cell lung cancer (NSCLC), many patients still experience disease recurrence and have limited treatment options. V940 (mRNA-4157) is an investigational individualized neoantigen therapy designed to train and activate an antitumor immune response based on the unique mutational signature of a patient's tumor (encoding up to 34 neoantigens). This study aims to demonstrate that combining V940 with pembrolizumab improves outcomes and provides a meaningful benefit over pembrolizumab alone in reducing disease recurrence.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled and Active Comparator-controlled (Pembrolizumab alone) **Blinding:** Double-blind **Randomization:** Randomized 1:1

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 868 evaluable patients **Number of Sites:** Global, multi-center trial (Locations include multiple sites across the US and internationally) **Estimated Study Start:** 06-Dec-2023 **Estimated Primary Completion:** 25-Jun-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years. 2. Completion of surgical resection of margin negative, completely resected non-small cell lung cancer (NSCLC), and has pathological Stage II, IIIA, IIIB (N2) squamous or nonsquamous tumor, node, metastasis (TNM) staging per American Joint Committee on Cancer (AJCC) Eighth Edition guidelines. 3. Prior treatment with at least one dose of adjuvant therapy with standard-of-care platinum-based doublet chemotherapy (up to four cycles). 4. No evidence of disease before randomization, and no more than 24 weeks have elapsed between surgical resection with curative intent and the first dose of pembrolizumab. 5. Participants who are hepatitis B surface antigen (HBsAg) positive must have received hepatitis B virus (HBV) antiviral therapy for at least 4 weeks and have undetectable HBV viral load prior to randomization. 6. Participants with a history of hepatitis C virus (HCV) infection must have an undetectable HCV viral load at screening. 7. Human immunodeficiency virus (HIV)-infected participants must have well-controlled HIV on anti-retroviral therapy (ART).

4.2 Exclusion Criteria 1. Diagnosis of small cell lung cancer (SCLC) or, for mixed tumors, presence of small cell elements, or has a neuroendocrine tumor with large cell components or a sarcomatoid carcinoma. 2. HIV-infected participants with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 3. Received prior neoadjuvant therapy for their current NSCLC diagnosis. 4. Received or is a candidate to receive radiotherapy for their current NSCLC diagnosis. 5. Received prior therapy with an anti-programmed cell death 1 protein (PD-1), anti-PD-ligand 1 (L1), or anti-PD-L2 agent, or with an agent directed to another stimulatory or coinhibitory T-cell receptor. 6. Received prior systemic anticancer therapy including investigational agents within 4 weeks before randomization. 7. Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. (Administration of killed vaccines is allowed). 8. Diagnosis of immunodeficiency or receiving chronic systemic steroid therapy (in dosing exceeding 10 mg daily of prednisone equivalent) or any other form of

immunosuppressive therapy within 7 days prior to the first dose of study medication. 9. Known additional malignancy that is progressing or has required active treatment within the past 5 years. 10. Active autoimmune disease that has required systemic treatment in the past 2 years (replacement therapy is allowed). 11. History of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease. 12. Active infection requiring systemic therapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants receive V940 (mRNA-4157) 1 mg every three weeks for up to nine doses, concurrently with Pembrolizumab 400 mg every six weeks for up to nine cycles. **Route of Administration:** V940 via intramuscular injection; Pembrolizumab via intravenous (IV) infusion. **Duration of Treatment:** Approximately one year, or until disease recurrence or other criteria for discontinuation are met.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for surgery and chemotherapy recovery; killed vaccines; replacement therapies (e.g., thyroxine, insulin, physiologic corticosteroids). **Prohibited:** Concurrent experimental or anti-cancer therapies, live or live-attenuated vaccines, and prior immunotherapy for NSCLC.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Post-Surgery/Chemo Informed Consent, Eligibility Confirmation, Tumor sample acquisition for V940 manufacturing Baseline Day 0 Randomization, First Dose of Pembrolizumab & V940 (or placebo), Baseline Imaging Interim Every 3 to 6 weeks Study drug administration, Safety Labs, Efficacy Assessment (Tumor imaging every 12 weeks until week 48) End of Study Year 1 Final treatment administration, Transition to follow-up and survival observation

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Disease-Free Survival (DFS), defined as the time from randomization to any recurrence (local, locoregional, regional or distant), occurrence of new primary NSCLC, or death due to any cause, whichever occurs first. **Measurement Method:** Assessed by the investigator using tumor imaging.

7.2 Secondary & Exploratory Endpoints 1. Overall Survival (OS) 2. Distant Metastasis-Free Survival (DMFS) 3. Lung Cancer Specific Survival (LCSS) 4. Quality of Life (QoL) metrics

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via clinical assessments, patient reporting, and laboratory tests throughout the treatment phase. **Serious Adverse Events (SAEs):** Standard 24-hour reporting window to the sponsor and regulatory authorities for all Grade 3/4 AEs or SAEs. **Data Monitoring Committee (DMC):** An independent DMC will review unblinded safety and efficacy data globally.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy analysis (DFS). Safety Set for all patients receiving at least one dose of the study drug. **Sample Size Justification:** $N = 868$ properly powered to evaluate the superiority of V940 + pembrolizumab compared to placebo + pembrolizumab. **Handling of Missing Data:** Standard time-to-event statistical methodologies (e.g., Kaplan-Meier estimation, Cox proportional hazards models) will be utilized for DFS and OS, incorporating standard censoring rules.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Continuous site monitoring (remote and onsite) to ensure protocol adherence, drug accountability, and patient safety. **Audit Trail:** Secure Electronic Case Report Forms (eCRF) with stringent audit trails and data clarification processes.

11. Study Identifiers & Governance

Primary ID: V940-002 **Registry Number:** NCT06077760 **Posting ID:** 350988239740467329 **Sponsor/Lead Organization:** Merck & Co., Inc. (MSD) & Moderna **Study Title:** INTERpath-002 **Official Regulatory Title:** A Phase 3, Randomized, Double-blind, Placebo- and Active-Comparator-Controlled Clinical Study of Adjuvant V940 (mRNA-4157) Plus Pembrolizumab Versus Adjuvant Placebo Plus Pembrolizumab in Participants With Resected Stage II, IIIA, IIIB (N2) Non-small Cell Lung Cancer

12. Clinical Framework (NSCLC Specific)

Primary Condition: Non-small Cell Lung Cancer (NSCLC) **Diagnosis Threshold:** Margin negative, completely resected Stage II, IIIA, IIIB (N2) **Medical Methodology:** Adjuvant Treatment with Individualized Neoantigen Therapy + PD-1 Inhibitor **Optimization Target:** Improvement of Disease-Free Survival (DFS)

13. Study Procedures & Methodology

Management Pathway: Adjuvant clinical pathway following definitive surgical resection and standard-of-care platinum-based doublet chemotherapy. **Education Protocol (HCP):** Site-specific training on V940 tumor sequencing workflows, handling of individualized mRNA therapeutics, and double-blind protocols. **Education Protocol (Patient):** Comprehensive informed consent covering personalized medicine, genomic sequencing of tumor samples, and visit frequency. **Quality Audit Mechanism:** Ongoing safety review and clinical site auditing.

14. Observation & Follow-Up Schedule

Total Duration: Primary completion estimated at ~7 years; maximum study follow-up up to 12 years (December 2035). **Onsite Visit Cadence:** Every 3 weeks for V940 (up to 9 doses) and every 6 weeks for Pembrolizumab (up to 9 cycles). **Remote Monitoring Frequency:** Routine oversight per monitoring charter. **Checklist Review Interval:** Tumor imaging occurs at baseline, every 12 weeks until week 48, every 24 weeks through year 3, and every 48 weeks thereafter.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				